Project Overview

Project 2: Environmental Analysis of Global Air Quality Data

Although invisible, the air we breathe is one of the most essential components of life. Oxygen sustains human survival; however, it also carries pollutants that can negatively affect human health. Activites such as infrastructure development, vehicle emissions, and industrial operations contribute to the increasing presence of harmful airborne particles. Monitoring and research play a crucial role in informing mitigation strategies and policies, particularly to protect vulnerable populations.

Air quality is commonly measured using the **Air Quality Index (AQI),** a standardized and color-coded system used by governments to communicate the level of air pollution. This index helps raise public awareness and enables individuals to take appropriate precautionary measures. AQI levels are categorized as *Good*, *Moderate*, *Unhealthy for Sensitive Groups*, *Unhealthy*, *Very Unhealthy*, and *Hazardous*.

This project provides a comprehensive analysis of a global air quality dataset, exploring how pollutant levels vary across cities and how environmental factors are associated with changes in air quality. The analysis aims to uncover patterns, identify high-risk conditions, and present insights that support better understanding of air pollution dynamics.

Objectives:
1. Conduct multiple analysis process on global air quality data.
2. Develop an interactive and informative dashboard.
3. Generate insights and conclusions based on observed patterns and relationships.

Dataset Summary:
1. The dataset contains hourly air quality measurements collected over a 15-day period across 50 major global cities from regions including *Asia*, *South America*, *North America*, *Oceania*, *Middle East*, *Europe*, and *Africa*.
2.Key variables include pollutant concentrations (*PM2.5, PM10,* NO_2, SO_2, *CO,* O_3), meteorological factors (*temperature, humidity, and wind speed*) and computed *AQI* values.

Data Source:
Author: Smeet Raichura
Title: Global Air Quality Data(15 Days Hourly, 50 Cities)
Link: https://www.kaggle.com/datasets/smeet888/global-air-quality-data15-days-hourly-50-cities
License: CC0: Public Domain

Note: *While this dataset represents air quality conditions across global cities, it is* ***synthetically generated*** *using historically consistent pollutant patterns and statistical ranges. It is designed to support research, modeling, experimentation, and analytical application without real-world data limitations.*

Section I: Data Foundation

Raw Air Quality Dataset

timestamp	country	city	latitude	longitude	pm25	pm10	no2	so2	o3	co	aqi	temperature	humidity	wind_speed
6:25:18 PM	US	New York	40.713	-74.006	50.295	108.938	27.998	6.539	52.568	1.096	108	18.504	70.168	3.725
7:25:18 PM	US	New York	40.713	-74.006	32.083	63.043	36.12	4.021	43.536	1.075	90	5.838	80.088	8.969
8:25:18 PM	US	New York	40.713	-74.006	42.25	82.553	26.935	9.538	23.32	0.977	84	31.833	62.783	9.65
9:25:18 PM	US	New York	40.713	-74.006	30.403	79.951	63.536	7.609	31.369	0.23	158	23.14	89.153	8.956
10:25:18 PM	US	New York	40.713	-74.006	21.083	66.423	38.997	6.919	45.615	1.085	97	13.632	76.499	4.017
11:25:18 PM	US	New York	40.713	-74.006	31.242	92.779	30.633	8.03	43.725	0.347	92	15.159	42.505	1.199
12:25:18 AM	US	New York	40.713	-74.006	77.69	65.198	20.302	7.641	62.687	0.734	155	36.729	47.651	4.542
1:25:18 AM	US	New York	40.713	-74.006	57.816	111.709	34.533	6.945	41.304	0.771	115	37.891	53.314	7.605
2:25:18 AM	US	New York	40.713	-74.006	60.914	44.775	40.936	6.779	36.751	1.014	121	24.092	48.148	8.686
3:25:18 AM	US	New York	40.713	-74.006	16.464	76.631	28.119	8.561	79.467	1.081	76	11.49	29.403	3.762
4:25:18 AM	US	New York	40.713	-74.006	48.469	72.858	23.356	4.244	34.794	0.973	96	27.913	34.736	5.054
5:25:18 AM	US	New York	40.713	-74.006	44.477	37.724	39.812	6.704	34	0.48	99	18.665	29.091	8.802
6:25:18 AM	US	New York	40.713	-74.006	57.227	28.916	30.698	8.091	63.108	0.967	114	14.584	46.654	5.57
7:25:18 AM	US	New York	40.713	-74.006	26.147	58.714	49.826	5.228	51.651	0.615	124	27.299	76.014	0.786
8:25:18 AM	US	New York	40.713	-74.006	38.815	121.394	25.969	4.112	44.73	1.124	121	34.481	81.79	3.576
9:25:18 AM	US	New York	40.713	-74.006	42.793	91.138	11.6	7.625	43.074	0.944	91	8.423	62.661	1.287
10:25:18 AM	US	New York	40.713	-74.006	2.799	69.074	52.218	5.538	37.461	0.6	130	7.884	74.873	3.401
11:25:18 AM	US	New York	40.713	-74.006	63.552	57.838	13.754	4.427	42.251	0.997	127	36.193	62.906	7.097
12:25:18 PM	US	New York	40.713	-74.006	26.664	54.577	35.136	3.599	68.139	1.069	87	14.656	83.829	5.372
1:25:18 PM	US	New York	40.713	-74.006	49.693	83.965	60.763	7.811	60.188	0.589	151	11.743	58.646	6.374
2:25:18 PM	US	New York	40.713	-74.006	22.22	102.306	37.332	5.142	32.491	0.781	102	12.79	65.872	7.227
3:25:18 PM	US	New York	40.713	-74.006	51.211	97.99	37.295	2.901	49.461	0.667	102	12.955	50.346	7.861
4:25:18 PM	US	New York	40.713	-74.006	18.385	27.829	19.647	6.642	54.364	0.818	49	32.209	25.117	8.398
5:25:18 PM	US	New York	40.713	-74.006	31.094	59.007	46.212	7.911	21.905	0.436	115	20.478	87.347	1.08
6:25:18 PM	US	New York	40.713	-74.006	38.005	25.92	23.383	4.869	42.954	0.884	76	14.901	30.594	4.689
7:25:18 PM	US	New York	40.713	-74.006	56.954	75.045	29.837	9.541	54.198	1.17	113	31.289	43.279	5.264

Section I: Data Foundation

Air Quality Dataset Table

timestamp datetime format	hour	day	timestamp original	country	city	region	latitude	longitude	pm25	p
04/11/2025 18:25	6:00 pm	Tue	6:25:18 PM	US	New York	North Americ	40.713	-74.006	50.295	
04/11/2025 19:25	7:00 pm	Tue	7:25:18 PM	US	New York	North Americ	40.713	-74.006	32.083	
04/11/2025 20:25	8:00 pm	Tue	8:25:18 PM	US	New York	North Americ	40.713	-74.006	42.25	
04/11/2025 21:25	9:00 pm	Tue	9:25:18 PM	US	New York	North Americ	40.713	-74.006	30.403	
04/11/2025 22:25	10:00 pm	Tue	10:25:18 PM	US	New York	North Americ	40.713	-74.006	21.083	
04/11/2025 23:25	11:00 pm	Tue	11:25:18 PM	US	New York	North Americ	40.713	-74.006	31.242	
05/11/2025 0:25	12:00 am	Wed	12:25:18 AM	US	New York	North Americ	40.713	-74.006	77.69	
05/11/2025 1:25	1:00 am	Wed	1:25:18 AM	US	New York	North Americ	40.713	-74.006	57.816	
05/11/2025 2:25	2:00 am	Wed	2:25:18 AM	US	New York	North Americ	40.713	-74.006	60.914	
05/11/2025 3:25	3:00 am	Wed	3:25:18 AM	US	New York	North Americ	40.713	-74.006	16.464	
05/11/2025 4:25	4:00 am	Wed	4:25:18 AM	US	New York	North Americ	40.713	-74.006	48.469	
05/11/2025 5:25	5:00 am	Wed	5:25:18 AM	US	New York	North Americ	40.713	-74.006	44.477	
05/11/2025 6:25	6:00 am	Wed	6:25:18 AM	US	New York	North Americ	40.713	-74.006	57.227	
05/11/2025 7:25	7:00 am	Wed	7:25:18 AM	US	New York	North Americ	40.713	-74.006	26.147	
05/11/2025 8:25	8:00 am	Wed	8:25:18 AM	US	New York	North Americ	40.713	-74.006	38.815	
05/11/2025 9:25	9:00 am	Wed	9:25:18 AM	US	New York	North Americ	40.713	-74.006	42.793	
05/11/2025 10:25	10:00 am	Wed	10:25:18 AM	US	New York	North Americ	40.713	-74.006	2.799	
05/11/2025 11:25	11:00 am	Wed	11:25:18 AM	US	New York	North Americ	40.713	-74.006	63.552	
05/11/2025 12:25	12:00 pm	Wed	12:25:18 PM	US	New York	North Americ	40.713	-74.006	26.664	
05/11/2025 13:25	1:00 pm	Wed	1:25:18 PM	US	New York	North Americ	40.713	-74.006	49.693	
05/11/2025 14:25	2:00 pm	Wed	2:25:18 PM	US	New York	North Americ	40.713	-74.006	22.22	
05/11/2025 15:25	3:00 pm	Wed	3:25:18 PM	US	New York	North Americ	40.713	-74.006	51.211	
05/11/2025 16:25	4:00 pm	Wed	4:25:18 PM	US	New York	North Americ	40.713	-74.006	18.385	
05/11/2025 17:25	5:00 pm	Wed	5:25:18 PM	US	New York	North Americ	40.713	-74.006	31.094	
05/11/2025 18:25	6:00 pm	Wed	6:25:18 PM	US	New York	North Americ	40.713	-74.006	38.005	
05/11/2025 19:25	7:00 pm	Wed	7:25:18 PM	US	New York	North Americ	40.713	-74.006	56.954	

Section I: Data Foundation

Calculations

Correlation

Variable	Correlation	Meaning
pm2.5	0.5561	moderate positive
no2	0.5317	moderate positive
pm10	0.2075	weak
o3	0.0080	negligible
temperature	0.0050	negligible
wind	0.0038	negligible
humidity	-0.0006	inverse relationship
co	-0.0018	inverse relationship
so2	-0.0052	inverse relationship

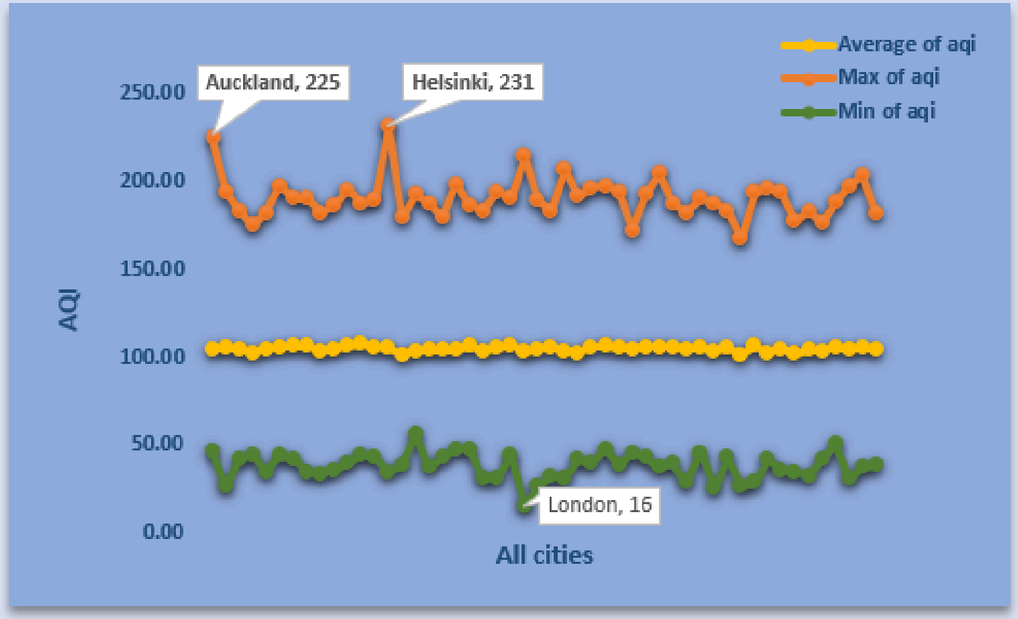
AQI by Region

City	Region	AQI
Auckland	Oceania	104.73
Bangkok	Asia	105.19
Beijing	Asia	104.41
Bengaluru	Asia	102.33
Berlin	Europe	104.42
Buenos Aires	South America	105.95
Cairo	Africa	106.05
Cape Town	Africa	106.16
Chennai	Asia	103.63
Chicago	North America	104.81
Delhi	Asia	106.44
Doha	Middle East	107.37
Dubai	Middle East	105.53
Helsinki	Europe	105.11
Ho Chi Minh City	Asia	101.54
Hong Kong	Asia	103.49
Houston	North America	104.09
Istanbul	Middle East	104.08
Jakarta	Asia	104.32
Karachi	Asia	106.30
Kolkata	Asia	103.53
Kuala Lumpur	Asia	105.01
Lagos	Africa	106.62
London	Europe	103.04
Los Angeles	North America	104.48

Section II: Analysis
Data Exploration

What is the average AQI for each country?
Which countries exhibit the highest and lowest AQI values?

Row Labels	Average of aqi	Max of aqi	Min of aqi
Auckland	104.73	225	47
Bangkok	105.19	194	27
Beijing	104.41	183	42
Bengaluru	102.33	175	45
Berlin	104.42	182	35
Buenos Aires	105.95	197	44
Cairo	106.05	190	42
Cape Town	106.16	190	35
Chennai	103.63	182	34
Chicago	104.81	186	36
Delhi	106.44	195	40
Doha	107.37	187	45
Dubai	105.53	189	43
Helsinki	105.11	231	35
Ho Chi Minh City	101.54	179	39
Hong Kong	103.49	192	56
Houston	104.09	187	38
Istanbul	104.08	179	43
Jakarta	104.32	198	48
Karachi	106.30	186	48
Kolkata	103.53	183	31
Kuala Lumpur	105.01	194	32
Lagos	106.62	190	44



Air Quality Index Category	
AQI	Category
0-50	Good
51-100	Moderate
101-150	Unhealthy for Sensitive Groups
150-200	Unhealthy
201-300	Very Unhealthy

Insights:

1. The average AQI across all monitored global cities is **104.65**, falling within the "**Unhealthy for Sensitive Groups**" category. This indicates that poor air quality is a widespread urban challenge rather than a localized issue.

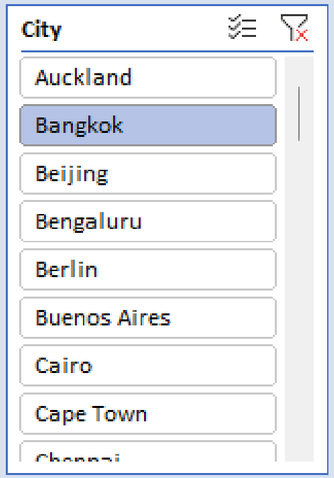
2. Among the cities analyzed, **Doha, Qatar** recorded the highest mean AQI at **107.37**, while **Shanghai, China** maintained the lowest mean at **101.38.**

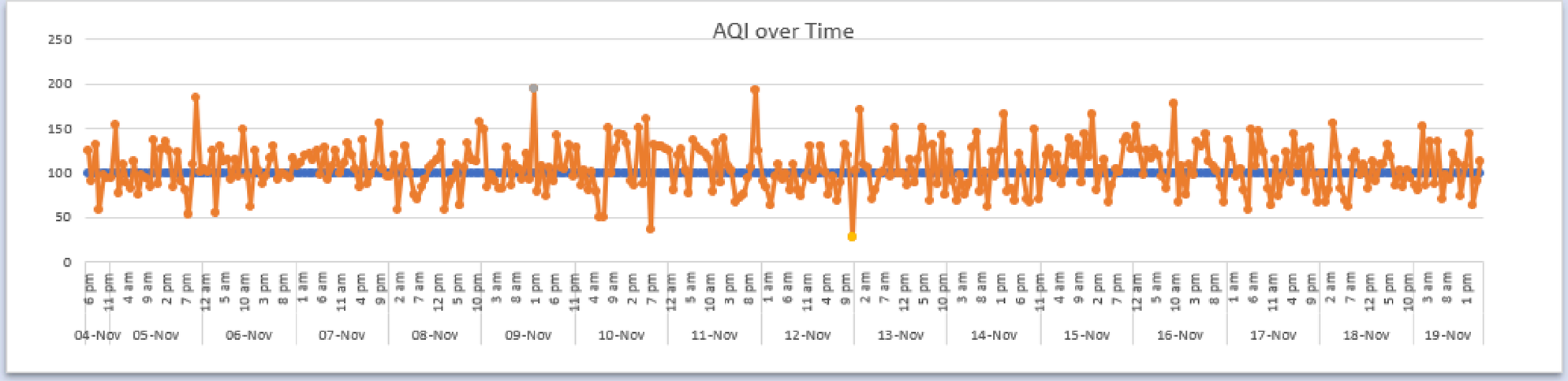
3. The most severe pollution event was recorded in **Helsinki, Finland,** with a peak AQI of **231 (Very Unhealthy**). Conversely, the cleanest air quality reading was observed in **London, Great Britain**, with a minimum AQI of **16 (Good).**

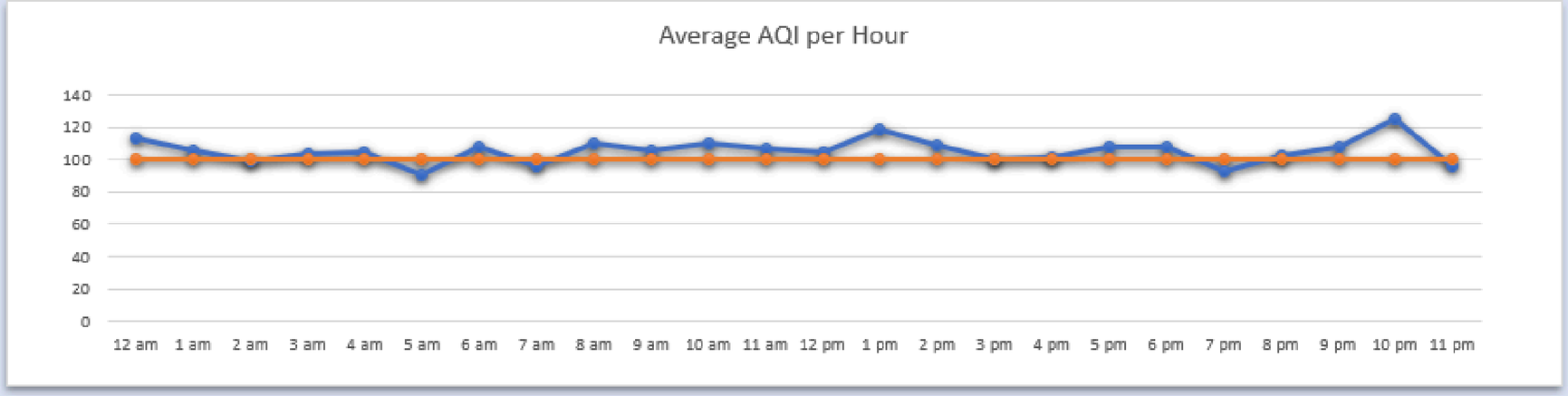
4. Although Finland is widely recognized for maintaining high air quality standards, the observed peak AQI value in Helsinki represents a deviation from expected conditions within the dataset. This suggest the influence of short-term or localized

Section II: Analysis
Time - Series with Threshold

How does AQI vary over the duration of the study period?
How does AQI behave across different hours of the day?







Fluctuations in the Air Quality Index (AQI) are clearly observed, with multiple readings exceeding the moderate threshold and reaching unhealthy levels, alongside periods indicating good air quality.

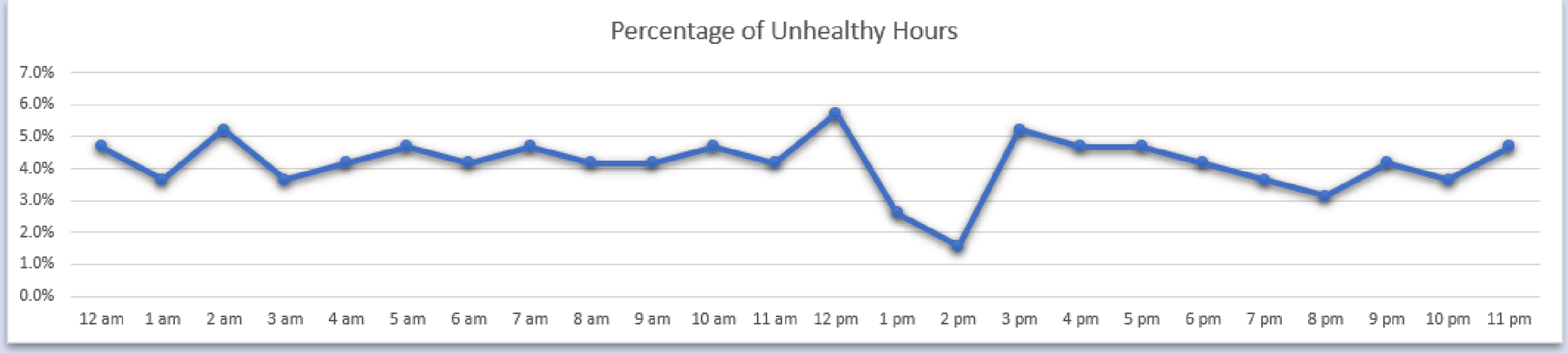
Variations in AQI across cities suggest the influence of multiple underlying factors, including pollutant concentrations and meteorological conditions.

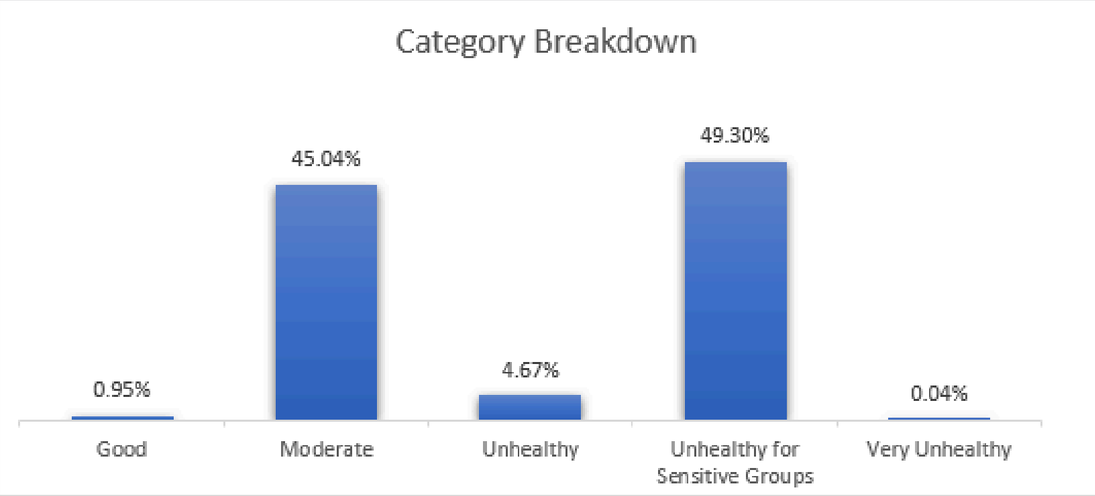
Section II: Analysis
Threshold

At what point does air quality exceed safe levels?
What proportion of observed hours fall under unhealthy conditions?

City

- Paris
- Riyadh
- Rome
- San Francisco
- Sao Paulo
- Seoul
- Shanghai
- Singapore
- Stockholm
- Sydney
- Tehran
- Tokyo
- Toronto
- Vancouver
- Warsaw
- Zurich
- SÃ£o Paulo





The chart highlights instances where average AQI values exceed established thresholds, consistent with patterns observed in the time series analysis. Variations across cities are also evident, with higher percentages of unhealthy air quality occurring during early morning hours and late afternoon periods.

The category distribution further indicates that good air quality represents the smallest proportion of observation, while the majority falls within moderate to unhealthy categories.

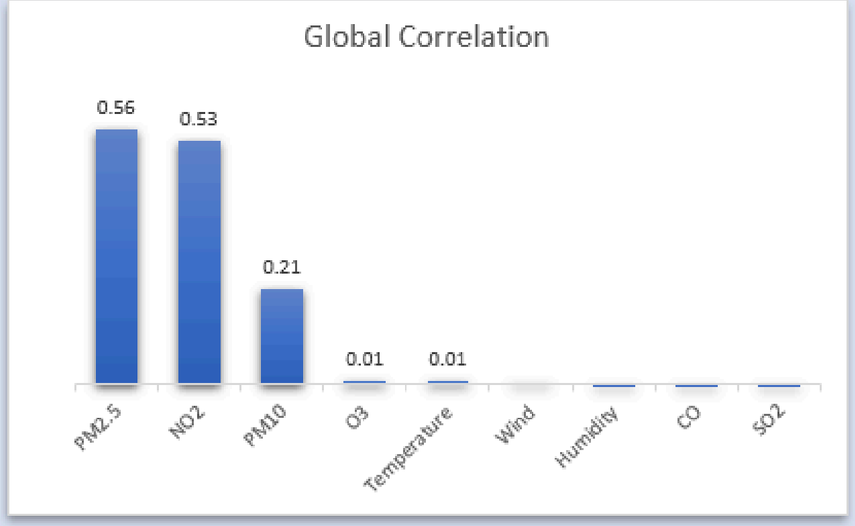
Section II: Analysis
Correlation

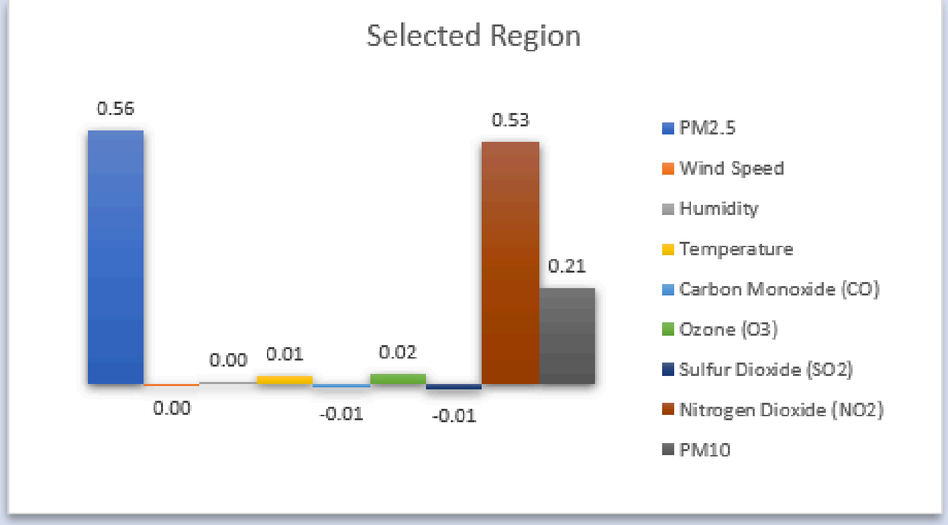
Which variables show the strongest association with AQI?

Variable	Correlation	Meaning
PM2.5	0.556094239	Moderate Positive
NO2	0.531730549	Moderate Positive
PM10	0.207499084	Weak
O3	0.008000164	Negligible
Temperature	0.004977975	Negligible
Wind	0.00379824	Negligible
Humidity	-0.000610308	Negligible (negative)
CO	-0.001762729	Negligible (negative)
SO2	-0.005201896	Negligible (negative)

Region

- Africa
- Asia
- Europe
- Middle East
- North America
- Oceania
- South America



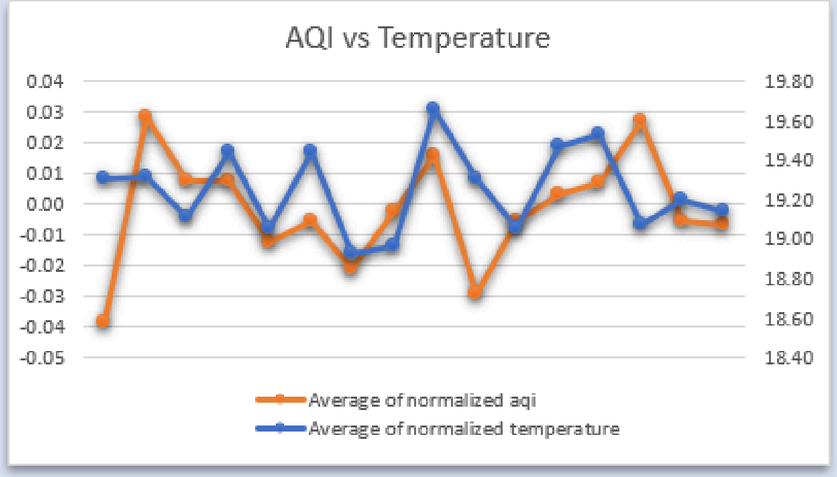


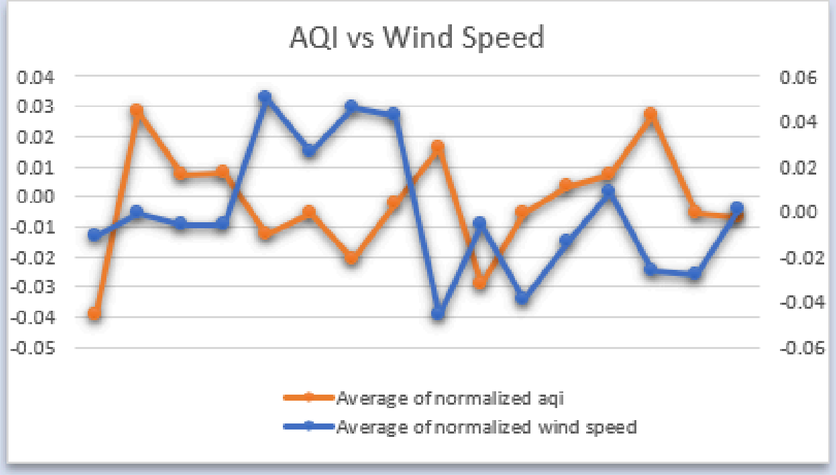
PM2.5 exhibits the strongest association with the Air Quality Index (AQI), showing a moderate positive correlation of **0.56**, followed by **nitrogen dioxide (NO_2)** at **0.53** and **PM10 at 0.21**. These pollutants are key contributors to air quality degradation. Particulate matter (PM2.5 and PM10) consists of fine inhalable particles originating from both human activities and natural processes, such as vehicle emissions, combustion, and construction. Similarly, nitrogen dioxide is primarily produced through combustion processes, particularly from fossil fuel usage. The consistency of these leading variables across regions suggest that air pollution is a widespread issue rather than a purely localized phenomenon.

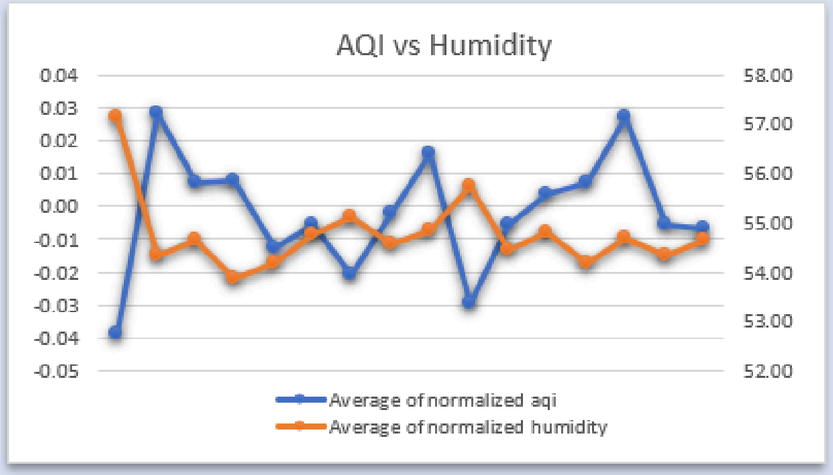
In contrast, meteorological factors exhibit relatively weak correlations with AQI. This indicates that while they may not directly determine pollutant levels, they influence the movement, dispersion, and accumulation of pollutants. Further analysis is therefore required to better understand their role in shaping air quality patterns.

Section II: Analysis
Variable Behavior

How do meteorological factors behave in relation to AQI levels?







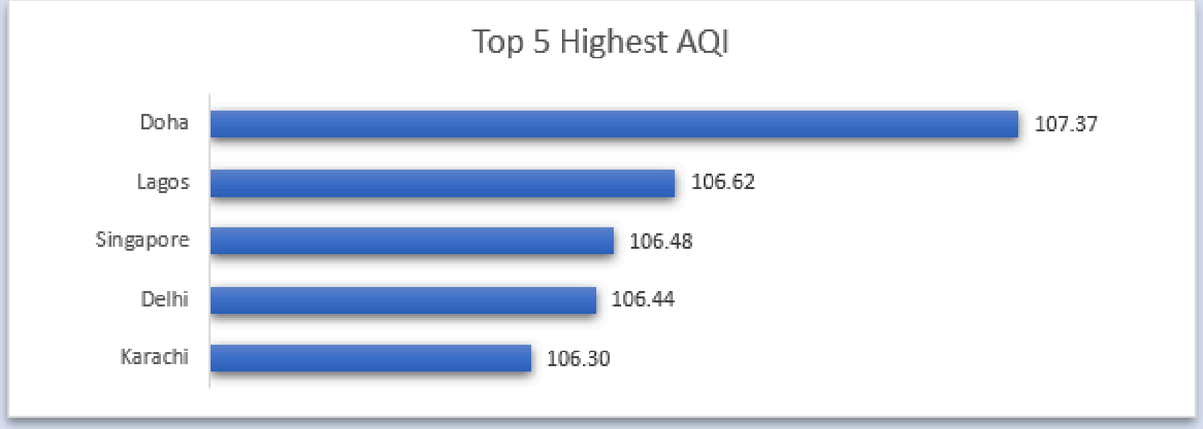
The variable behavior analysis indicates that meteorological factors exhibit varying degrees of association with AQI. In the first graph, the behavior of temperature appears to influence fluctuations in AQI. As temperature increases, AQI also shows a tendency to rise, suggesting that temperature may contribute to changes in air quality conditions. The second graph illustrates the relationship between wind speed and AQI. An opposing pattern is observed, where increases in wind speed are associated with decreases in AQI, and vice versa. This indicates an inverse relationship, which is consistent with the role of wind in dispersing pollutants and reducing their accumulation in a given area. Lastly, the graph for humidity shows inconsistent behavior. Initial observations suggest an inverse relationship with AQI; however, later readings indicate a weaker or unclear association. This variability suggest that humidity's influence on air quality is complex and may be affected by other factors such as atmospheric conditions or seasonal variations.

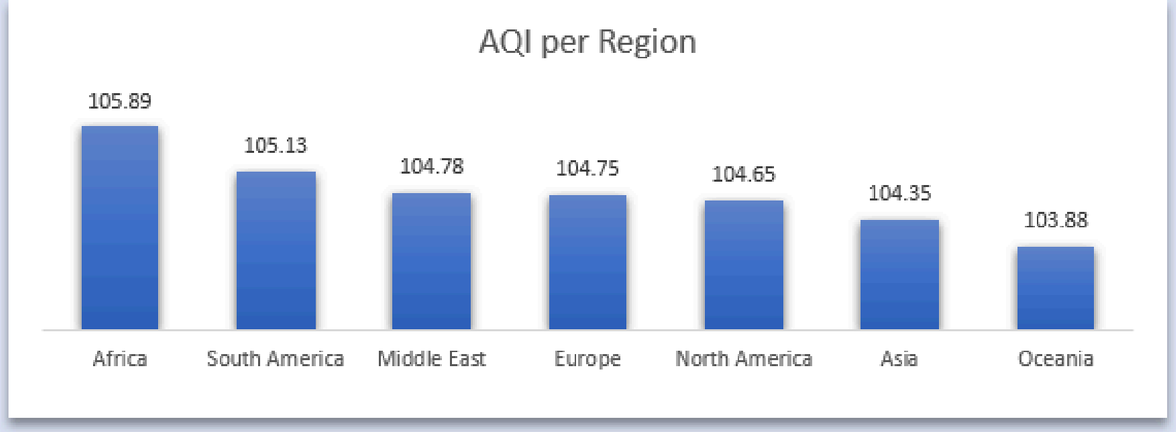
Overall, while meteorological variables do not directly detemine AQI levels, they influence the environmental conditions that govern pollutant behavior, highlighting the importance of considering both emission sources and atmospheric dynamics in air quality analysis.

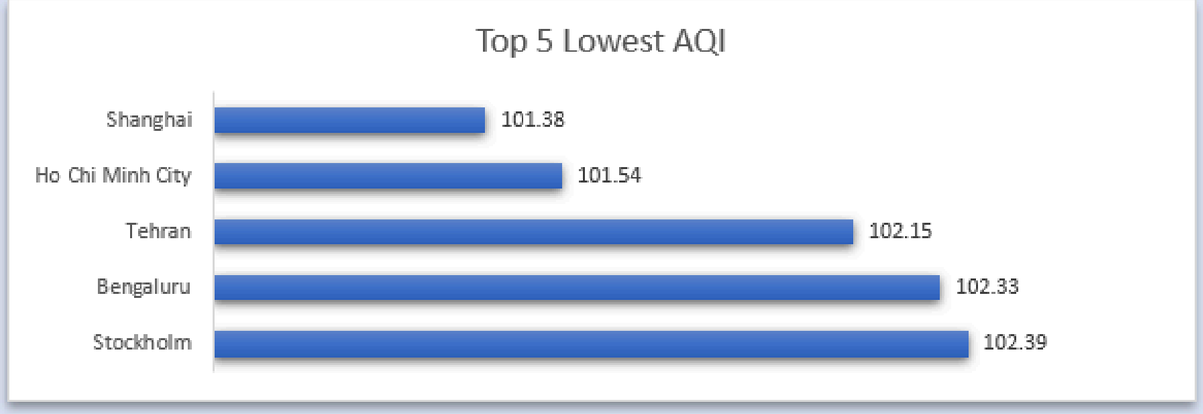
Section II: Analysis
Comparative

Chart Area elation" Point "Pm10"
Value: 0.207499084
Legend

Which cities exhibit the highest and lowest average AQI levels?
How does air quality vary across regions based on average AQI?







The city with the highest average AQI is **Doha, Qatar**, while **Shanghai, China** recods the lowest average AQI among the cities analyzed.

At the regional level, **Africa** exhibits the highest overall AQI, whereas **Oceania** shows the lowest, with a difference of 2.01 AQI units, indicating comparatively cleaner air conditions.

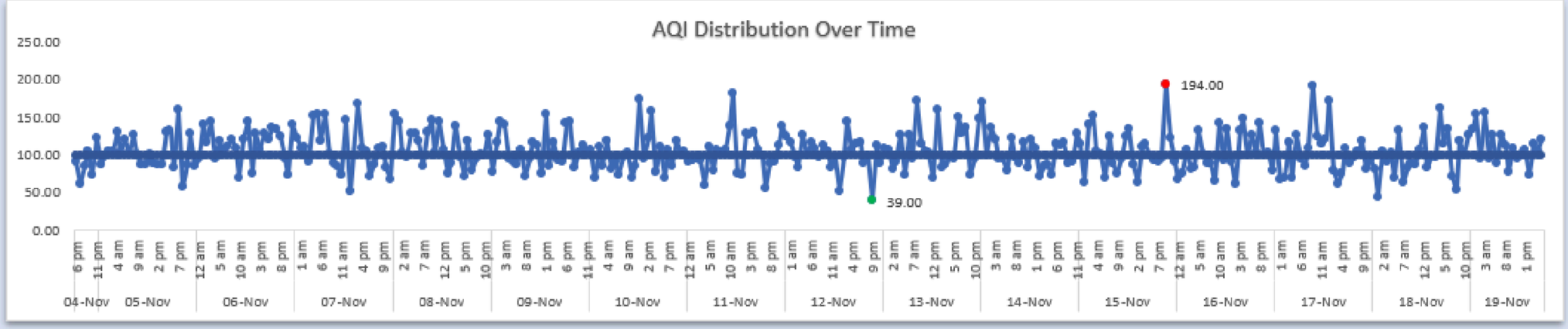
These differences highlight the variability of air quality across regions, which may be influenced by factors such as urbanization, industrial activity, and environmental policies. Notably, all observed cities and regions exceed the moderate air quality threshold, indicating that global air quality during the 15-day period falls within the "**Unhealthy for Sensitive Groups**" category. This emphasizes the importance of implementing effective strategies to address air pollution.

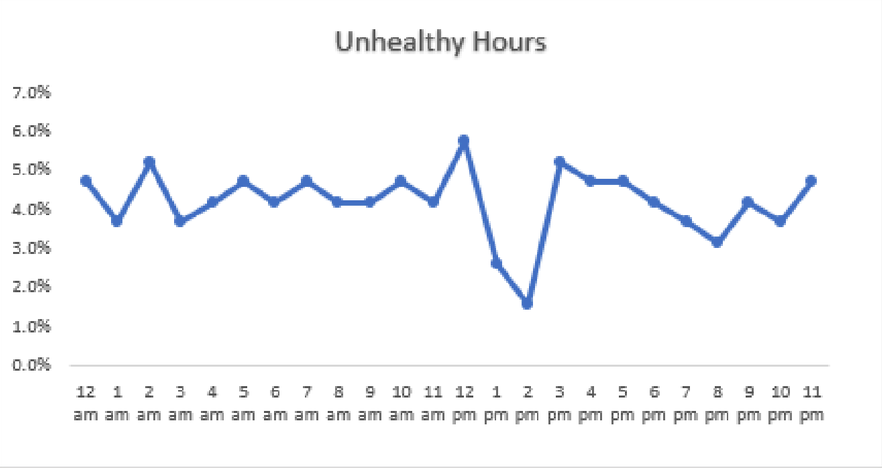
Global Air Quality Insights Dashboard | Mumbai

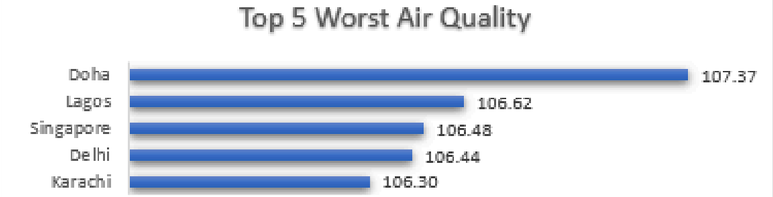
Average AQI	Dominant Category	Unhealthy Hours	City Rank	Region
105.40	USG	12:00 PM, 3:00 PM, 12:00 AM Top 3	33rd From Lowest AQI	Asia

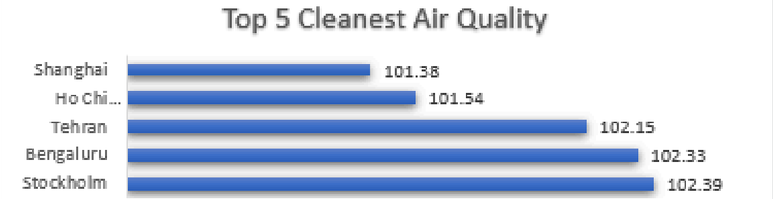
City

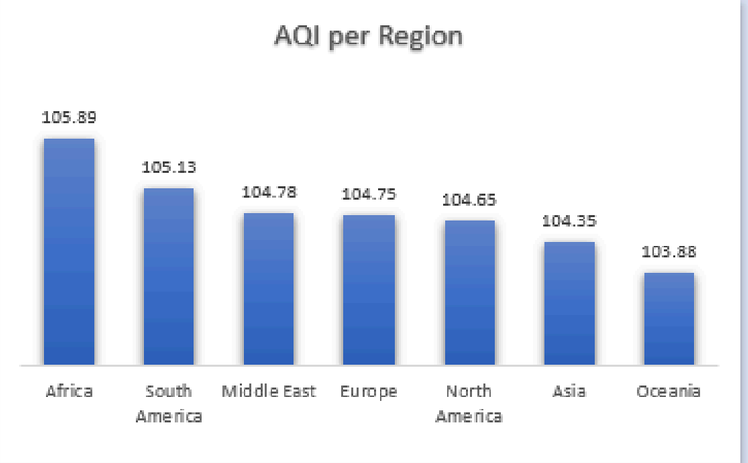
- Auckland
- Bangkok
- Beijing
- Bengaluru
- Berlin
- Buenos Aires
- Cairo











Created by: Joahna Rose P. Catubay

Overall Insight Report

Someone once told me that breathing fresh air is a luxury. Growing up in the province and later moving to the city for college, I found myself agreeing with that idea. During my daily commute to the university, I often had to wear a protective mask to avoid inhaling dust particles. This became even more important when I conducted my third-year mini-research on air quality, where I learned that PM2.5 particles can enter the bloodstream and negatively affects a person's health. Because of this experience, I chose air quality as the focus of my second project, as it is something I am familiar with and interested in exploring further.

The dataset used in this project was sourced from **Kaggle** and contains over **18,000** data points from **50** major global cities. It was synthetically generated for research and modeling purposes and represents hourly air quality measurements over a **15-day** period. I began by checking the structure of the data and preparing it for analysis before proceeding to the different stages of the project.

From the analysis, several general observations were made. In the data exploration stage, most cities exceeded the moderate air quality category and fell under the "**Unhealthy for Sensitive Groups**" classification. The time series analysis showed noticeable fluctuations in AQI across cities, with several instances going beyond acceptable thresholds. Further threshold analysis revealed that unhealthy air conditions occured frequently, while **good** air quality had the smallest proportion among the categories. In the correlation analysis, **PM2.5**, **nitrogen dioxide**, and **PM10** appeared to be the variables most associated with increases in AQI. On the other hand, meteorological factors such as temperature, wind speed, and humidity showed weaker relationships but still influenced how pollutants behaved and spread. The comparative analysis also showed differences across regions, with **Africa** having the highest AQI and **Oceania** the lowest.

Overall, the results suggest that air pollution is not just a localized concern but a global issue. While I am still learning about environmental monitoring, this project helped me better understand how different factors contribute to air quality. It also highlights the importance of continuous monitoring, research, and practical strategies in addressing air pollution.